



The \$19 cleaning cloth?

Apple starts selling its own microfiber cloth with embossed Apple logo on it for \$19 bucks. <https://digit.in/nov21-62>



PS5 surpasses the Switch

PS5 becomes the first console to outsell the Nintendo Switch in close to three years. <https://digit.in/nov21-63>

Time to move to DDR5?

With DDR5 memory now available, does it make sense to upgrade from DDR4 to DDR5? Agent001 investigates.

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ntel is bringing a bunch of goodies to the consumer PC space along with their 12th Gen 'Alder

Lake' Core processors. The most notable of these happen to be PCIe Gen 5.0 standard and DDR5 memory standard. While PCIe Gen 3.0 is far from being properly utilised by graphics cards, the introduction of PCIe Gen 4.0 felt a bit premature. However, NVMe solid storage drives really made the upgrade to PCIe Gen 4.0 worth it. PCIe Gen 5.0 might have a similar impact but only time will tell. As for now, it seems like DDR5 will have more of an impact.

DDR5 improves upon DDR4 across several aspects such as speeds, power efficiency and memory capacity. Alder Lake Z690 platforms still have a capacity limit of 128 GB which is the same as DDR4. Only enterprise servers will get to take advantage of the higher capacities that DDR5 has to offer. Then there's the power efficiency improvement which translates to about 8.4% reduced power consumption over DDR4. That's not much but you need to realise that the PMIC (Power Management IC) which handles the power levels and therefore, the overclocking controls, lies on the DIMMs instead of the processor. So the modules need to be supplied with more power to

compensate for the additional power control circuitry on the DIMMs. For desktops, this power efficiency doesn't matter much but on laptops and portable devices, this means you get better battery life.

On the performance front, there are several improvements. The PMIC on the DIMMs ensures better overclocking capabilities for enthu-



Fresh off the boat

siasts. Then there's the bandwidth improvement which is a huge bump over DDR4. DDR4 modules are generally seen with data rates of 2133-3600 MT/s and the overclocked modules can go to upwards of 4200 MT/s. DDR5 starts off from 4800 MT/s and the stable sweet spot is around 5200 MT/s. Think of this as 2x the performance over DDR4. And this is without overclocking. We've already seen overclockers push DDR5

modules over 8000 MT/s showing the tremendous overhead DDR5 has in terms of overclocking potential. So DDR5 has a massive performance potential. The question is, are there any applications that can benefit from this potential performance?

Since only the Intel Z690 platforms are DDR5 compatible, that's the only use case we can look at and

if we are to compare DDR4 vs DDR5 performance across a wide variety of use cases, then some interesting results pop out. Synthetic tests such as AIDA64 which writes to and reads from the RAM directly shows a tremendous difference in performance between DDR4 and DDR5 memory

standards. However, a real-world test such as extracting or compressing a massive set of files shows a slightly reduced performance gap. And if you're transcoding videos from one format to another then you're going to see some decent saving in workload times as well.

Then there's gaming. Since there are a lot of different game engines out there with different levels of optimisations, you can never tell how